

L1 — Equivalence Completion (Paper 12): Exhaustion and Closure of the Equivalence Space

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0. Scope

This paper proves exhaustion. It introduces no new primitives, no new ontology, no new anchors, and no new tensor load. It establishes the non-extendability of L1 under admissibility, standing, and closure.

1. Apparent Incompleteness

After closing multiple equivalences, one may suspect that further admissible equivalences remain. This paper shows that such suspicion is structurally impossible.

2. Exhaustive Failure Classes

Across Papers 1–11, all admissible standing failures are exhausted: redundancy non-exhaustion, non-transportability, overcounting, illicit continuity, and compensating structure dependence. No additional admissible failure class exists.

3. Exhaustion Argument

Assume a further admissible equivalence exists. Then it must arise from an apparent freedom that avoids all known failure classes. No such freedom exists without introducing new primitives or violating admissibility. Hence the assumption is false.

4. Core Result

Theorem 1 (Equivalence Exhaustion). *Under admissibility and closure, the space of Standard-Model equivalences is exhausted by Papers L1–1 through L1–11. No further admissible equivalence completion is possible.*

Corollary 1 (Non-Extendability). *Any extension of L1 necessarily violates admissibility or reintroduces already-eliminated failure modes.*

Corollary 2 (Global Uniqueness). *The equivalence structure of the Standard Model is unique up to admissible redescription.*

5. Closure Statement

L1 is closed.